Hypogonadotropic Hypogonadism 23 With or Without Anosmia (HH23) — Comprehensive Disease Report

Disease: Hypogonadotropic Hypogonadism 23 With or Without Anosmia (HH23) **Causal gene:** *SMCHD1* (Structural Maintenance of Chromosomes flexible Hinge Domain containing 1) **MONDO:** MONDO:0009223 · **Category:** Mendelian, autosomal dominant **Evidence base:** Human clinical case reports/series, in vitro biochemistry, mouse/Xenopus models, and human iPSC systems. No patient-level dataset was provided; this report is synthesized from aggregated disease-level resources and primary literature.

Summary (Answer to the Research Question)

HH23 is a rare Mendelian form of congenital hypogonadotropic hypogonadism (GnRH deficiency), with or without anosmia, caused by heterozygous variants in the epigenetic regulator **SMCHD1**. It lies at the hypogonadism-predominant end of the **SMCHD1 allelic spectrum** whose severe extreme is **Bosma arhinia microphthalmia syndrome (BAMS; OMIM 603457)** — characterized by arhinia (absent nose), anosmia with absent olfactory bulbs, microphthalmia/coloboma, hypogonadotropic hypogonadism and cryptorchidism, with normal cognition
(
P 28067911

P 26842768

).

P 38808953

Disease-causing variants cluster in the **N-terminal extended GHKL-type ATPase domain** and behave as gain/altered-function alleles, in contrast to the loss-of-function *SMCHD1* alleles that cause facioscapulohumeral muscular dystrophy type 2 (FSHD2)
(
P 28067911

). P 28222895 The hypogonadism component is managed as for congenital hypogonadotropic hypogonadism generally — sex-steroid replacement and gonadotropin/pulsatile-GnRH therapy for fertility (P 26194704). P 27467188

1. Disease Information

Overview. HH23 is a congenital GnRH-deficiency disorder (central/secondary hypogonadism) presenting with absent or incomplete puberty and infertility, which may be accompanied by impaired olfaction (anosmia/hyposmia) — the Kallmann-syndrome phenotype — and, at its most severe, by the craniofacial malformation complex of BAMS. It is one of the numbered loci (PS147950 phenotypic series) of congenital hypogonadotropic hypogonadism (CHH), attributed to *SMCHD1*.

Key identifiers. - **MONDO:** MONDO:0009223 - **Gene:** *SMCHD1*, HGNC:29090, NCBI Gene 23347, UniProt A6NHR9, chromosome 18p11.32 - **Related OMIM phenotype:** Bosma arhinia microphthalmia syndrome, #603457 (allelic/overlapping) - **Orphanet (BAMS/arhinia):** ORPHA:1134 (arhinia); **MeSH:** Kallmann Syndrome (D017436); Hypogonadism (D007006) - **ICD-11:** 5A61.0 (hypogonadotropic hypogonadism); **ICD-10:** E23.0 (hypopituitarism/Kallmann)

Synonyms / alternative names. HH23; HH23 with or without anosmia; *SMCHD1*-related hypogonadotropic hypogonadism; isolated GnRH deficiency (*SMCHD1*); within the broader entity Bosma arhinia microphthalmia syndrome (BAMS, Bosma syndrome). Kallmann syndrome denotes the anosmic subtype.

Data derivation. Disease-level, derived from aggregated literature (OMIM, Orphanet, PubMed case reports and reviews), not from individual EHR data.

2. Etiology

Causal factor — genetic. Heterozygous **germline** missense variants in *SMCHD1*, predominantly **de novo** (P 28067911).
The condition is not infectious or primarily environmental.

Genetic risk factors. - Causal variants map to the N-terminal **extended ATPase (GHKL) domain** (exons ~3–11). Reported examples: p.Glu473Val (c.1418A>T; with hypogonadotropic hypogonadism), p.His348Arg (c.1043A>G, de novo), p.Asp398Asn (isolated HH/CPHD spectrum) (P 38808953 P 33784779). P 32620854
- **Modifier:** for the FSHD2 phenotype, the 4q35 D4Z4 repeat length and permissive 4qA haplotype act as strong modifiers; arhinia patients can co-carry an FSHD2-permissive background and show DUX4 derepression (P 35121673).
This epigenetic/genetic context likely contributes to the striking phenotypic divergence.

Environmental risk factors. None established. Age, sex, and specific exposures are not known contributors; disease is congenital and genetically determined.

Protective factors. No genetic or environmental protective factors are documented. (gnomAD: constrained gene; pathogenic ATPase-domain missense variants are absent/ultra-rare in controls.)

Gene–environment interactions. None demonstrated. The relevant "interaction" is intrinsic gene–epigenome: the pathogenic effect depends on *SMCHD1*'s chromatin context and, for the muscle phenotype, the D4Z4 locus.

3. Phenotypes (with HPO terms, frequency, onset, severity)

Phenotype	HPO term	Type	Onset	Frequency/ notes
Hypogonadotropic hypogonadism	HP:0000044	Lab/clinical (endocrine)	Congenital; recognized at puberty (or neonatally as micropenis/ cryptorchidism)	Defining; present across the spectrum
Absent/delayed puberty	HP:0000823 / HP:0000826	Clinical sign	Adolescence	Common in untreated patients
Anosmia/hyposmia	HP:0000458	Symptom	Congenital	In the "with anosmia" (Kallmann) subset; variable
Absent/hypoplastic olfactory bulbs	HP:0007686	Imaging sign	Congenital	Correlates with anosmia on MRI
Arhinia / absent nose	HP:0004122 (abnormal nasal morphology); arhinia	Physical malformation	Congenital	Severe (BAMS) end
Choanal atresia	HP:0000453	Physical malformation	Congenital	BAMS
Microphthalmia	HP:0000568	Physical malformation	Congenital	BAMS
Coloboma	HP:0000589	Physical malformation	Congenital	BAMS
High-arched palate / cleft palate	HP:0000218 / HP:0000175	Physical malformation	Congenital	BAMS
Cryptorchidism	HP:0000028	Physical sign	Congenital	Males
Micropenis	HP:0000054	Physical sign	Congenital/ neonatal	Males

Phenotype	HPO term	Type	Onset	Frequency/ notes
Normal intelligence	HP:0001522 (absent) — cognition normal	—	—	Characteristic; distinguishes from midline brain malformations

- **Severity:** variable — from isolated hypogonadotropic hypogonadism to full BAMS.
- **Progression:** congenital and **stable/non-progressive** structurally; the endocrine deficiency is lifelong unless the ~10–20% reversal occurs (P 26194704).
- **Quality-of-life impact:** infertility, delayed/absent sexual development and associated psychosocial burden; anosmia impairs flavor perception and hazard detection; arhinia severely affects breathing/feeding, facial appearance, and lacrimal drainage (P 40788678). Cognition and life expectancy are typically preserved in mild forms (P 26842768).

Evidence:

P 26842768

P 38808953

P 33784779

P 32620854 .

4. Genetic / Molecular Information

- **Causal gene:** *SMCHD1* (HGNC:29090; OMIM gene 614982), 18p11.32; encodes a ~2,005-aa non-canonical SMC-family protein with an **N-terminal GHKL-type ATPase** (plus ubiquitin-

like domain required for N-terminal dimerization/chromatin localization) and a **C-terminal SMC hinge** that homodimerizes and binds nucleic acid (P 34109974, P 32546545, P 32779700).

- **Variant type/class:** predominantly **missense**, clustered in the extended ATPase domain (P 28067911).
- **Classification:** pathogenic / likely pathogenic by ACMG/AMP (de novo occurrence PS2; domain clustering/absence from controls); some rare variants remain VUS (e.g., p.Asp398Asn classified likely pathogenic, P 32620854).
- **Allele frequency:** pathogenic variants are absent or ultra-rare in gnomAD; *SMCHD1* is intolerant to such missense in the ATPase region.
- **Origin:** germline; de novo in most BAMS/HH cases (P 28067911). No somatic/cancer role in this disease.
- **Functional consequence:** BAMS/HH ATPase-domain variants behave as **gain/altered-function** (Xenopus assays), the opposite directionality to **loss-of-function** FSHD2 alleles (P 28067911, P 28222895).
- **Modifier genes:** 4q D4Z4 repeat/haplotype (muscle phenotype); other CHH/Kallmann genes may act oligogenically.
- **Epigenetic information:** *SMCHD1* is itself an epigenetic effector; patients carry a reproducible DNA-methylation "episignature," and FSHD2-overlap cases show D4Z4 hypomethylation with DUX4 derepression (P 37334829, P 35121673).

- **Chromosomal abnormalities:** none characteristic; disease arises from point mutations, not aneuploidy/translocation.

---

5. Environmental Information

- **Environmental factors / toxins / radiation:** none implicated.
- **Lifestyle factors:** none causal (relevant only to general reproductive/bone health in management).
- **Infectious agents:** none. This is a non-infectious congenital genetic disorder.

(Section not applicable beyond stating absence of environmental/infectious etiology.)

---

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **de novo heterozygous missense variant** in the *SMCHD1* N-terminal extended ATPase (GHKL) domain **alters** SMCHD1 enzymatic/chromatin activity (gain/altered-function; demonstrated biochemically and in Xenopus)
   (P 28067911).
2. Altered SMCHD1 **perturbs** its normal role in long-range chromatin compaction and heterochromatin maintenance — recruitment to targets via the Xist–HnrnpK–PRC1/H2AK119ub pathway and deposition/maintenance of H3K9me3 and H3K27me3 blocks and CpG-island methylation
   (P 30428357 P 35831949 P 30126901 P 18425126).

(Mechanism at targets well demonstrated; the precise mis-silenced genes driving human nasal/HH phenotypes are inferred.)

3. Dysregulated silencing **leads to** mis-expression of developmental master regulators (e.g., Hox clusters, clustered protocadherins, imprinted/neural-crest fate genes) in euchromatin and repressed chromatin of **neural crest stem cells** and craniofacial/olfactory placode progenitors
(P 37334829).
(Inferred effector loci.)
4. This **results in** failed morphogenesis of the **nasal placode/olfactory system** → arhinia and absent/hypoplastic olfactory bulbs (branch A) and disrupted **craniofacial/ocular** development → microphthalmia, coloboma, palatal/choanal defects (branch B)
(P 28067911, P 26842768).
5. Branch C — **reproductive axis:** absent/defective olfactory bulb scaffolding and/or independent hypothalamic effects **impair embryonic migration of GnRH-synthesizing neurons** from the olfactory placode to the hypothalamus (the canonical Kallmann mechanism), and/or GnRH-neuron function, **leading to** deficient hypothalamic GnRH secretion
(P 26194704).
(For SMCHD1 specifically, the GnRH-neuron step is inferred by analogy to Kallmann biology rather than directly demonstrated.)
6. GnRH deficiency **causes** low pituitary LH/FSH secretion → **results in** low gonadal sex-steroid output (testosterone/estradiol) and impaired gametogenesis → **manifests as** hypogonadotropic hypogonadism, absent puberty, cryptorchidism/micropenis, and infertility
(P 26194704, P 28833369).
7. Chronic untreated hypogonadism **leads to** downstream osteoporosis and metabolic syndrome
(P 30352392).

Supporting detail

- **Molecular pathways:** epigenetic silencing (Polycomb PRC1/PRC2, H2AK119ub → H3K27me3; H3K9me3 heterochromatin), 3D genome organization (TAD/compartment regulation on Xi). Downstream, the reproductive phenotype funnels into the **hypothalamic–pituitary–gonadal (GnRH→LH/FSH→sex steroid) axis** (P 30604745). P 26194704
- **Cellular processes:** chromatin compaction, dosage compensation, cell-fate determination/differentiation of neural crest; **GnRH neuron migration** (olfactory placode → hypothalamus).
- **Protein dysfunction:** altered ATPase/dimerization behavior of mutant SMCHD1; UBL domain needed for N-terminal dimerization and chromatin localization (P 34109974).
- **Immune/metabolic:** no primary immune involvement; metabolic syndrome is a late consequence of hypogonadism.
- **GO terms:** GO:0031507 heterochromatin formation; GO:0040029 epigenetic regulation of gene expression; GO:0009048 dosage compensation by inactivation of X chromosome; GO:0006346 methylation-dependent chromatin silencing; GO:0001945/GO:0097475 neuron migration (GnRH neuron migration); GO:0042755 (not applicable). **CL terms:** CL:0000333 neural crest cell; CL:0011111 GnRH neuron; CL:0000207 olfactory receptor neuron; CL:0000216 gonadotroph (Sertoli/Leydig downstream).

---

7. Anatomical Structures Affected

- **Organ level (primary):** hypothalamus (UBERON:0001898) and pituitary gland (UBERON:0000007) — reproductive axis; olfactory system (olfactory bulb UBERON:0002264, olfactory epithelium UBERON:0001997); external/internal nose (UBERON:0000004) in BAMS; eye (UBERON:0000970); gonads — testis (UBERON:0000473)/ovary (UBERON:0000992).
- **Secondary involvement:** bone (osteoporosis), cardiovascular/metabolic (metabolic syndrome), lacrimal system (in arhinia); airway/respiratory compromise neonatally (arhinia/choanal atresia).

- **Body systems:** endocrine (reproductive), nervous (olfactory), craniofacial/skeletal, ocular, respiratory (secondary).
- **Tissue/cell level:** neural crest–derived craniofacial mesenchyme; olfactory placode epithelium; GnRH neurons; gonadal germ/somatic cells (CL:0000333, CL:0011111, CL:0000207).
- **Subcellular (GO Cellular Component):** nucleus (GO:0005634), chromatin/heterochromatin (GO:0000785/GO:0000792), inactive X chromosome (Barr body, GO:0001739).
- **Lateralization:** malformations are usually **bilateral** (arhinia, olfactory bulb aplasia, microphthalmia) but **hemiarhinia (unilateral)** occurs as a milder form (P 38808953). P 40788678

---

8. Temporal Development

- **Onset: congenital.** Structural anomalies present at birth (arhinia, microphthalmia, choanal atresia); hypogonadism is congenital but often first recognized at expected puberty (or neonatally via micropenis/cryptorchidism/absent mini-puberty).
- **Onset pattern:** chronic/insidious for the endocrine component; the nasal/airway malformation is an acute neonatal emergency in severe cases.
- **Progression: non-progressive/stable** structurally; endocrine deficiency is **lifelong/chronic** barring the ~10–20% spontaneous reversal seen in CHH generally (P 26194704).
- **Critical periods:** embryonic craniofacial/olfactory morphogenesis and GnRH-neuron migration (first trimester) for the malformations; **mini-puberty (0–6 months)** and expected pubertal window for diagnosis and hormonal intervention; neonatal period for airway management.
- **Remission:** treatment-induced puberty/fertility; a minority show spontaneous recovery of reproductive function.

---

9. Inheritance and Population

- **Epidemiology:** ultra-rare. BAMS has <**50 reported cases/series**; arhinia overall <**30–40 documented worldwide**
  (P 26842768). P 41832654
  No reliable prevalence/incidence figures exist; it is far rarer than Kallmann syndrome overall (~1/10,000 males, ~1/50,000 females for CHH/KS; P 30352392).
- **Inheritance: autosomal dominant**, heterozygous *SMCHD1* variants on 18p11; **predominantly de novo**
  (P 38808953). P 28067911
- **Penetrance/expressivity: incomplete penetrance** and **markedly variable expressivity** (isolated HH → hemiarhinia → full BAMS; and distinct FSHD2 for LoF alleles)
  (P 35121673). P 28222895
- **Anticipation:** not a repeat-expansion disorder; no anticipation.
- **Germline mosaicism:** possible (relevant to recurrence counseling); not well quantified.
- **Founder effects / consanguinity / carrier frequency:** none established; de novo origin makes carrier screening non-applicable.
- **Demographics:** reported across diverse ethnicities and geographies with no clustering; both sexes affected (males more readily recognized via genital findings; BAMS diagnostic criteria emphasize HH in males). No defined sex ratio.

10. Diagnostics

Clinical/biochemical. - Reproductive hormone panel: low/low-normal **LH** and **FSH** with low **testosterone/estradiol**; low **inhibin B** and **AMH**; **GnRH and hCG stimulation** tests to assess axis integrity (P 28833369). LOINC-type analytes: LH, FSH, testosterone, estradiol, inhibin B. - Evaluate other pituitary axes to exclude **combined pituitary hormone deficiency / septo-optic dysplasia** overlap (P 32620854).

Olfactory testing. Objective and subjective **olfactometry**; results correlate with olfactory-bulb volume on MRI (P 25553765).

Imaging. Dedicated **MRI of the olfactory system** (bulb/sulcus aplasia or hypoplasia; may show anterior pituitary hypoplasia) (P 26571055); P 25553765 craniofacial **CT/MRI** for arhinia, choanal atresia, microphthalmia/coloboma; **fetal MRI/ultrasound** can detect arhinia prenatally (P 39295111).

Genetic testing. Molecular diagnosis = ***SMCHD1* sequencing**, best delivered within a **multigene CHH/Kallmann panel** or **WES/WGS**, given genetic heterogeneity (P 41902607). P 32620854 Chromosomal microarray/karyotype/FISH are generally **not** diagnostic (point mutations). For patients with a *SMCHD1* variant, consider FSHD2 risk assessment (4q D4Z4 methylation/haplotype).

Clinical criteria. Proposed BAMS criteria: **arhinia, hypoplastic maxilla, normal cognition, and hypogonadotropic hypogonadism (in males)** (P 26842768).

Differential diagnosis. Other Kallmann/CHH genes (ANOS1, FGFR1, FGF8, PROK2/PROKR2, CHD7, SOX10, GNRHR, KISS1/KISS1R, etc.); CHARGE syndrome; septo-optic dysplasia; holoprosencephaly (for arhinia); constitutional delay of puberty (P 26194704). P 41902607

Screening. No population/newborn screening. Cascade testing has low yield (de novo); prenatal/PGT applies when a familial variant is known.

11. Outcome / Prognosis

- **Survival/life expectancy:** normal in isolated/mild HH23; **neonatal airway obstruction** from arhinia/choanal atresia is the chief early-life mortality risk in severe BAMS, often requiring intubation/tracheostomy (P 33784779).
- **Cognition:** normal intelligence and brain structure (P 26842768).
- **Morbidity:** untreated hypogonadism → **infertility, osteoporosis, metabolic syndrome** (P 30352392); anosmia (safety/flavor); ocular impairment from microphthalmia; facial disfigurement/airway/lacrimal issues from arhinia.
- **Recovery/treatment response:** fertility inducible in most CHH with gonadotropins/GnRH; normal final testicular volume attained in ~70–74% and sperm in ejaculate in 91–95% with combined gonadotropin therapy; ~10–20% spontaneous reversal

(P 27467188). P 26194704

- **Prognostic factors:** completeness of GnRH deficiency, severity of arhinia/microphthalmia, timeliness of hormone replacement and neonatal airway/surgical management. No molecular prognostic biomarker beyond genotype/phenotype correlation.

---

12. Treatment

Pharmacotherapy (hypogonadism). - **Sex-steroid replacement** to induce/maintain puberty and secondary sex characteristics and protect bone: **testosterone** (males; NCIT:C2270, CHEBI:17347) or **estrogen/progesterone** (females; estradiol CHEBI:16469). Does not restore fertility. - **Fertility induction: hCG** (NCIT:C1947) + **recombinant FSH/hMG** (FSH NCIT:C105867), or **pulsatile GnRH** (gonadorelin; NCIT:C76341, CHEBI:5520). Restores testicular growth and spermatogenesis in the large majority; ovulation induction in females (P 27467188 P 33894111 P 30519672 P 40463510). P 26194704

- **Infancy:** gonadotropin therapy during mini-puberty can increase penile length and inhibin B and promote testicular descent (P 39673783).

Surgical/interventional (BAMS features). Staged **nasal reconstruction** (e.g., costal-cartilage framework with forehead/expander flaps), **Le Fort II** midface procedures, choanal atresia repair, ophthalmologic management, and **dacryocystorhinostomy** for lacrimal obstruction (P 41902607

P 41832654

).

P 40788678

Neonatal airway stabilization (oral airway/tracheostomy).

Advanced/experimental. No gene-, cell-, RNA-based, or SMCHD1-targeted therapy exists for HH23. No disease-specific clinical trials; management follows CHH guidelines (

P 26194704

).

Pharmacogenomics / personalized medicine. None specific; treatment is genotype-agnostic for the endocrine axis (same across CHH genes).

Outcomes/adverse events: gonadotropin/testosterone therapies are generally well tolerated; monitor erythrocytosis, acne, gynecomastia, injection burden; surgical risks per procedure.

Strategy. Multidisciplinary: endocrinology (hormone replacement/fertility), ENT/craniofacial & plastic surgery, ophthalmology, genetics/counseling, psychology.

---

13. Prevention

- **Primary prevention:** not feasible (predominantly de novo genetic disorder).
- **Secondary prevention / early detection:** prenatal fetal MRI/ultrasound can detect arhinia (

  P 39295111

  ); early recognition of micropenis/cryptorchidism/absent mini-puberty enables timely endocrine diagnosis.
- **Tertiary prevention:** timely sex-steroid and gonadotropin therapy to prevent osteoporosis, metabolic syndrome, and infertility (

  P 30352392

  );

  P 26194704

  bone-density monitoring; anticipatory **neonatal airway planning** for arhinia (

  P 33784779

).

- **Genetic counseling:** explain de novo origin (low but non-zero recurrence from possible gonadal mosaicism), ~50% transmission from an affected reproducing proband (AD), and variable expressivity/incomplete penetrance; offer **prenatal/preimplantation genetic testing** when a pathogenic *SMCHD1* variant is identified. NCIT:C15681 (Genetic Counseling), NCIT:C15417 (Prenatal Diagnosis).
- **Immunization / public-health / environmental measures:** not applicable.

---

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *Homo sapiens* (NCBI Taxon 9606), *SMCHD1* (Gene 23347); *Mus musculus* (10090), *Smchd1* (Gene 74355); *Xenopus laevis* (8355). *Smchd1* is evolutionarily conserved; the related plant GHKL ATPase (DMS11) participates in RNA-directed DNA methylation, underscoring deep conservation of GHKL-ATPase epigenetic function (P 22560611).
- **Natural disease / veterinary relevance:** no naturally occurring *SMCHD1*-arhinia/HH disorder is documented in companion animals or wildlife (no OMIA entry reported here). Not zoonotic; no cross-species transmission.
- **Comparative biology:** mouse *Smchd1* loss affects X-inactivation and autosomal (Hox/imprinted) silencing, conserving the human molecular role, but does not reproduce arhinia or hypogonadism.

---

15. Model Organisms

- **Mouse (*Smchd1*):** the *MommeD1* ENU allele / *Smchd1*-null causes **female-specific mid-gestation lethality** with failed X-inactivation maintenance and X-linked gene derepression (P 18425126); *Smchd1*-deficient epiblast stem cells/MEFs show loss of Xi H3K9me3/H3K27me3 and altered chromosome architecture (P 35831949)

P 30604745

).

P 30126901

Resources: MGI, IMPC/KOMP, IMSR.

- **Xenopus laevis:** human BAMS/HH ATPase-domain variants expressed in embryos behave as **gain/altered-function** alleles affecting development — key evidence for variant directionality
( ).
P 28067911
- **Human cellular / iPSC:** iPSC-derived **neural crest stem cells** and FSHD2/BAMS primary cells define SMCHD1 **episignatures** and tissue-specific gene-regulatory effects
( ).
P 37334829
- **Phenotype recapitulation / limitations:** models capture the **epigenetic silencing (FSHD2/LoF) biology** well but **do not reproduce arhinia or hypogonadotropic hypogonadism** (the gain/altered-function BAMS/HH phenotype); constitutive null is female-lethal. No validated mouse or zebrafish model of HH23 arhinia/HH is established, so developmental mechanism is studied mainly in Xenopus and human iPSC systems.

Supported vs. Refuted Hypotheses

Supported. 1. HH23 is caused by heterozygous *SMCHD1* ATPase-domain variants
(
P 28067911

P 38808953

).
P 32620854

2. *SMCHD1* is a non-canonical SMC epigenetic silencer; BAMS/HH variants are gain/altered-function, opposite to FSHD2 LoF
(
P 28222895

).
P 28067911

3. The phenotype is congenital, AD, predominantly de novo, with variable expressivity and

incomplete penetrance (P 28067911 P 38808953 P 35121673). 4. Hypogonadism is treatable with sex steroids + gonadotropins/GnRH (P 26194704 P 27467188).

Refuted / not supported. - No evidence for environmental, infectious, or lifestyle causation. - No founder effect, sex-linked inheritance, anticipation, or somatic/cancer mechanism. - Arhinia+IHH is **not** exclusively *SMCHD1* — other genes (e.g., GNRHR/KISS1) can produce a similar presentation (P 41902607), so genotype must be confirmed.

Limitations and Future Directions

- Ultra-rarity → no registries, prevalence/incidence estimates, penetrance quantification, or QoL instrument data specific to HH23.
- The **direct molecular link** from mutant SMCHD1 to GnRH-neuron/olfactory failure is **inferred**, not demonstrated; mis-silenced effector loci driving the human nasal/HH phenotype are unknown.
- No faithful animal model of the gain-of-function BAMS/HH phenotype; needed for mechanism and therapeutics.
- Genotype–phenotype rules (why the same domain yields isolated HH vs. full BAMS) remain unresolved; epigenetic/modifier context is a priority research area.

Evidence types: human clinical (case reports/series, cohort/consensus), in vitro biochemistry/ structural biology, mouse and Xenopus models, and human iPSC systems, as cited by PMID throughout.